mdstats FINAL-GPU1 Workstation Qualification Runbook

mdstats development

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CURRENT TARGET-SIZE-v5 HANDOFF

This runbook describes the current one-shot FINAL-GPU1 handoff for the target-size-v5 software generation. It is intentionally independent of the retired SIZE-FIDELITY2/MVMIGRATE1 activation workflow. Target-size v5 is already the production architecture; FINAL-GPU1 qualifies accelerator execution and release performance only.

1. Inputs

Prepare one workstation directory containing:

- the exact release source archive being qualified;
- the locked `mace-mh-1.model` and `mace-mpa-0-medium.model` files;
- the target/replay data required by the release-matched campaign;
- the project offline dependency bundle; and
- a writable qualification root.

Do not replace either foundation model. The tool verifies both model SHA-256 identities against the locked `mdstats` constants.

2. Runtime prerequisite

Activate the CUDA environment intended to authorize the release. It must contain the release-matched MACE/e3nn stack plus the CuEquivariance core, Torch frontend, and CUDA operations package appropriate to the installed PyTorch/CUDA runtime.

Missing accelerator capability is a failed/deferred runtime, not permission to alter scientific settings or silently fall back inside the same qualification record.

3. Verify and unpack the release

Verify the source archive with the checksum manifest supplied with the final bundle, then unpack it without editing source. The exact archive SHA-256 is part of every handoff registration.

Example:

```
sha256sum -c VERIFY_BUNDLE_SHA256.txt
mkdir -p work/source
unzip -q <release-archive.zip> -d work/source
cd work/source/<release-directory>
```

4. Preflight

```
python tools/run_mlff_final_gpu_qualification.py preflight \
  --mh1-model ../../../inputs/mace-mh-1.model \
  --mpa0-model ../../../inputs/mace-mpa-0-medium.model \
  --release-archive ../../../<release-archive.zip> \
  --output ../../../final-gpu1/preflight.json
```

The current preflight is the target-size-v5 **v11** contract. A `ready_for_final_gpu_execution` result requires the locked models, a readable release archive, CUDA availability, and a positive CUEQ-DEP1 runtime freeze.

5. Initialize one immutable handoff root

```
python tools/run_mlff_final_gpu_qualification.py init \
  --root ../../../final-gpu1/run-001 \
  --mh1-model ../../../inputs/mace-mh-1.model \
```

```
  --mpa0-model ../../../inputs/mace-mpa-0-medium.model \
  --release-archive ../../../<release-archive.zip>
```

The current policy creates a 16-item matrix: 8 `must_pass`, 6 `measure_only`, and 2 `optional` gates. Do not re-run `init` on a populated root. Replacement evidence belongs in a new run root.

6. Capture CUEQ-DEP1 once

```
mkdir -p ../../../final-gpu1/run-001/raw
python tools/capture_mlff_cueq_dep1_runtime.py \
  --supplied-artifact ../../../inputs/mace-mh-1.model \
  --supplied-artifact ../../../inputs/mace-mpa-0-medium.model \
  --output ../../../final-gpu1/run-001/raw/cueq_dep1_runtime.json

python tools/run_mlff_final_gpu_qualification.py record \
  --root ../../../final-gpu1/run-001 \
  --gate CUEQ_DEP1_RUNTIME_FREEZE \
  --evidence ../../../final-gpu1/run-001/raw/cueq_dep1_runtime.json \
  --disposition auto
```

All runtime-bound evidence must authenticate to this exact runtime digest.

7. Execute the current matrix

7.1 Must-pass gates

Gate ID	Required evidence
`CUEQ_DEP1_RUNTIME_FREEZE`	positive release-authorizing CuEq/CUDA runtime freeze
`E3NN_BASELINE_COMPLETE_CAMPAIGN`	complete optimized e3nn production-representative baseline
`SIZE_FIDELITY1_EXHAUSTIVE_CALIBRATION`	release qualification of the current size-screen calibration policy
`PERF_P2R_WHOLE_FUNNEL_GPU_PERFORMANCE`	whole-funnel GPU/control-plane performance with restart/decision equivalence
`VRAM1_PERF_P4_ACCELERATOR_MEMORY_THROUGHPUT`	VRAM/headroom/OOM/backoff and bounded-pipeline throughput evidence
`CUEQ_PHASE1_TRAINING_ONLY_QUALIFICATION`	paired e3nn/CuEq training qualification on one frozen runtime
`REPLAY_UNIFY1_GPU_PSEUDOLABEL_EXECUTION`	release-authoritative replay pseudolabel execution and cache/restart evidence
`PERF_CERT1_END_TO_END_CERTIFICATION`	end-to-end certification against the authoritative e3nn baseline

7.2 Measure-only gates

Register evidence for:

- `PREC3_REAL_CUEQ_ACTIVATION`
- `MH1_ACCEL1_CUEQ_NUMERICAL_PARITY`
- `MH1_DATA6_1_CUEQ_DESCRIPTOR_SELECTION_PARITY`
- `MH1_TRAIN1_CUEQ_TRAINING_REALIZATION`
- `MH1_CERT1_GENERATED_DEFAULT_CUEQ_MATRIX`
- `PERF_P5_ACCELERATOR_PERSISTENCE_REUSE`

These measurements must be complete even if a particular optimization remains disabled.

7.3 Optional gates

- `CUEQ_PHASE2_SELECTED_HEAD_SOURCE_EXECUTION_OPTIONAL`
- `MH1_DEPLOY1_MLIAP_EXPORT_AND_LAMMPS_RUN0`

8. Run the scientific campaign normally

Use the normal staged CLI. Do not bypass target-size selection or held-out boundaries:

```
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> doctor
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> prepare
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> preflight
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> train
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> evaluate
python tools/mdstats-mlff-campaign.py --config <frozen-campaign.toml> verify
```

The campaign itself owns the configured target-size ladder and fidelity path:

```
pi_train -> configured candidate ladder -> screen(n1/M1 -> n2/M2 -> n3/M3)
        -> N_selected and T_selected = pi_train[:N_selected]
        -> post-selection cross-validation on exactly T_selected
        -> fresh final production on the complete T_selected
```

FINAL-GPU1 must not create rescue sizes, migrate old target ladders, or alter a selected target size.

9. Register evidence

For every matrix item, use the handoff tool rather than editing the manifest:

```
python tools/run_mlff_final_gpu_qualification.py record \
  --root ../../../final-gpu1/run-001 \
  --gate <GATE_ID> \
  --evidence ../../../final-gpu1/run-001/raw/<evidence.json> \
  --disposition auto
```

The registration binds the evidence bytes, schema/content digest where available, release archive, and CuEq runtime digest where required.

10. Verify integrity before reduction

```
python tools/run_mlff_final_gpu_qualification.py verify \
  --root ../../../final-gpu1/run-001
```

Any source/model/evidence mutation, policy/matrix drift, path escape, missing runtime binding, or inconsistent producer status fails closed.

11. Reduce FINAL-GPU1

```
python tools/run_mlff_final_gpu_qualification.py reduce \
  --root ../../../final-gpu1/run-001 \
  --runtime ../../../final-gpu1/run-001/raw/cueq_dep1_runtime.json \
  --perf-cert1 ../../../final-gpu1/run-001/raw/perf_cert1_qualification.json \
  --output ../../../final-gpu1/run-001/FINAL_GPU1_QUALIFICATION.json
```

A positive record requires every must-pass gate to pass, every measure-only gate to be complete, handoff integrity to pass, and all runtime/release/model identities to agree.

A pass may recommend an accelerator profile but does not directly change generated defaults. The handoff record keeps generated-default authorization false.

12. Retired target-size migration workflow

Do **not** run retired migration, rescue, or target-data activation commands. Those belonged to historical campaign generations and are not prerequisites for the current target-size architecture. Current campaigns reject retired derived target-size state before reuse and rebuild the target-size authority from the current P1/P2/P3 owners.